

Vibrational Deduction Transformer (VDT)

A Spectral-PPCA Variational Autoencoder with Graph-Wiring Latent Space and Associative Spectral Memory

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Abstract

We introduce the *Vibrational Deduction Transformer* (VDT), a generative architecture that treats the latent space of a Variational Autoencoder (VAE) as a vibrational system grounded in Rayleigh’s *Theory of Sound*. The VDT decoder maps a stochastic latent code $\mathbf{z}$ to a *spectral loading matrix* $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$, where $\mathbf{U}_{1:q}$ are the q lowest-frequency eigenvectors of a frozen ArrowSpace graph-Laplacian index $\mathbf{L}(\mathcal{I})$, $\mathbf{S}$ is a learnable coefficient matrix in that eigenbasis, and $\boldsymbol{\omega}$ are mode weights drawn from a taumode variational prior. Here $\mathbf{W} \in \mathbb{R}^{N \times q}$ where N is the number of graph nodes and q is the number of retained spectral modes. This loading matrix parametrises a differentiable Laplacian $\mathbf{L}(\mathbf{z})$ over which a heat-kernel *taumode diffusion* decoder reconstructs the input. The objective is a four-term Evidence Lower Bound (ELBO) comprising a Gaussian reconstruction term, an isotropic latent KL, an eigenvalue-weighted spectral-basis KL penalising high-frequency loadings (spectral Occam’s razor), and a Gamma-vs-Exponential mode-frequency KL for spectral band selection. After training, the model emits a *spectral artefact* $\mathcal{A}(\mathcal{I}) = \{\hat{\mathbf{W}}, \hat{\boldsymbol{\omega}}, \mathbf{S}_{\text{mem}}\}$ that initialises a Hopfield-type associative memory for a downstream transformer. Benchmarks on CORA, PubMed, and synthetic spring-network graphs demonstrate entropy-controlled spectral graph generation and strong linear-probe accuracy of frozen latent representations.

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1 Introduction

Transformer-based generative models typically organise learned representations in a flat isotropic Gaussian latent space, discarding the spectral structure of the data. With ArrowSpace we demonstrated that any vector space can span a semantic web of relations from its feature-space via graph reconstruction algorithm (Graph Wiring); this structure carries critical semantic information that a flat prior cannot capture.

The Vibrational Deduction Transformer (VDT) is motivated by three converging lines of inquiry:

1. **Rayleigh’s vibrational analogy.** Rayleigh [1] showed that the natural oscillation modes of a continuous medium are the eigensolutions of its governing differential operator. Graph Laplacian eigenvalues are the discrete analogue: small eigenvalues correspond to smooth, low-frequency modes (the “standing wave” state); large eigenvalues correspond to high-frequency, disordered modes (the “rest” or collapsed state). The VDT treats the *low-frequency oscillation state as the low-entropy prior* and high-frequency disorder as high entropy, directly reversing the usual thermodynamic framing: the natural state of the string is to oscillate, not to be still.
2. **Real-valued quantum amplitudes.** Aaronson [2] observes in Chapter 9 of *Quantum Computing Since Democritus* that the essential features of quantum probability—interference, entanglement structure, the Born rule—are recoverable from signed real amplitudes without complex numbers, provided the update rule is unitary. The VDT implements this insight: the entries of the loading matrix $\mathbf{W}$ can be positive or negative, modulating the relative magnitude of Laplacian eigenvector contributions and producing constructive or destructive interference in the heat kernel through the eigenvector structure of $\mathbf{L}(\mathbf{z})$.
3. **ArrowSpace / Graph Wiring.** The ArrowSpace computational framework [3] defines a differentiable Laplacian builder and a spectral cost (J_{freq}) that penalises high-frequency wiring. The VDT replaces this hard penalty with a *variational* counterpart—a KL divergence over spectral loadings and mode weights—yielding a proper Bayesian ELBO.

The paper follows the conceptual flow applied in [5].

The paper is organised as follows. Section 2 reviews prerequisite material. Section 3 presents the full VDT architecture. Section 4 derives the four-term ELBO. Section 5 describes the spectral artefact and associative memory. Section 6 covers the stability analysis. Section 7 reports empirical results. Section 8 discusses related work. Section 9 concludes.

2 Background

2.1 Graph Laplacians and Spectral Graph Theory

Given an undirected weighted graph $G = (V, E, w)$ with $N = |V|$ nodes, the *combinatorial Laplacian* is $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ is the degree matrix and $\mathbf{A}$ the weighted adjacency. The normalised Laplacian $\tilde{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$ has eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \leq 2$. The eigenvectors $\{\mathbf{u}_k\}$ form an orthonormal basis (the *graph Fourier basis*), and the *Rayleigh quotient*

$$\mathcal{R}(\mathbf{v}) = \frac{\mathbf{v}^\top \mathbf{L} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}} \quad (1)$$

gives the frequency (smoothness cost) of signal $\mathbf{v}$ on the graph [6]. This is the discrete analogue of Rayleigh’s classical frequency functional for continuous media [1].

2.2 Probabilistic PCA and Variational Autoencoders

Probabilistic PCA (PPCA) [7] provides a generative model $p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$ where $\mathbf{W} \in \mathbb{R}^{D \times q}$ is the loading matrix, D is the input feature dimension, and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Variational Autoencoders (VAEs) [8] extend this to nonlinear decoders by maximising the Evidence Lower Bound $\mathcal{L}_{\text{VDT}} = \mathbb{E}[\log p(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$. The reparameterisation trick $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, enables gradient-based optimisation.

2.3 Real Quantum Amplitudes and Signed Interference

Aaronson [2] finds plausible that the observable consequences of quantum mechanics—interference, entanglement structure, Born-rule probabilities—can be derived from the assumption of signed real probability amplitudes, provided the time-evolution operator is a *real orthogonal* (rather than complex unitary) matrix. The key insight is:

“If we allow *negative* probability amplitudes, then we get quantum-like interference without complex numbers.” [2]

The VDT operationalises this at the level of the *loading matrix*: entries of $\mathbf{W} \in \mathbb{R}^{N \times q}$ can be positive or negative, modulating the relative magnitude and sign of each eigenvector’s contribution to the row embeddings of $\mathbf{L}(\mathbf{z})$. Edge weights themselves remain strictly positive (Section 3.4); the signed character of $\mathbf{W}$ instead reshapes the eigenvector decomposition of $\mathbf{L}(\mathbf{z})$, inducing constructive or destructive interference in the heat kernel $\mathbf{K}_\tau = \mathbf{U} \exp(-t\boldsymbol{\Lambda}) \mathbf{U}^\top$ through the geometry of the learned eigenvectors.

3 Architecture

3.1 Overview

The VDT is a *Spectral-PPCA VAE* whose generative path is mediated by a learned graph wiring constrained to the eigenbasis of an ArrowSpace index $\mathcal{I}$. Figure 1 summarises the data flow.

3.2 WiringEncoder

The encoder is a multi-layer perceptron (MLP) augmented with a *λ -fingerprint*—a fixed scalar summary of the Laplacian spectrum that encodes the global spectral geometry of the graph:

$$\text{fingerprint}(\mathbf{L}(\mathcal{I})) = f(\lambda_1, \dots, \lambda_q) \in \mathbb{R}^{d_{\text{fp}}}, \quad (2)$$

where f is a deterministic aggregation function (e.g. normalised eigenvalue histogram). The encoder produces:

$$\mathbf{z}, \boldsymbol{\mu}, \log \sigma^2, \log \mathbf{a}, \log \mathbf{b} = \text{WiringEncoder}(\mathbf{x}, \mathbf{L}(\mathcal{I})), \quad (3)$$

where $\mathbf{z} \in \mathbb{R}^{d_z}$ is the reparameterised VAE latent, $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}_{>0}^q \times \mathbb{R}_{>0}^q$ are the shape and rate parameters of the Gamma variational posterior over mode weights.

3.3 SpectralLoadingDecoder

A linear projection $\mathbf{z}_q = \mathbf{W}_{z \rightarrow q} \mathbf{z} \in \mathbb{R}^q$ bridges the VAE latent dimension d_z to the spectral mode dimension q . The spectral loading decoder then produces:

$$\mathbf{S} = \text{S_net}(\mathbf{z}_q) \in \mathbb{R}^{q \times q}, \quad \text{posterior mean of spectral coefficients}, \quad (4)$$

$$\log \sigma_{\mathbf{S}}^2 = \log_var_S_head(\mathbf{z}_q) \in \mathbb{R}^{q \times q}, \quad \text{independent posterior log-variance}, \quad (5)$$

$$\boldsymbol{\omega} = \exp(\text{omega_net}(\mathbf{z}_q)) \in \mathbb{R}_{>0}^q, \quad \text{mode weights}, \quad (6)$$

$$\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S} \in \mathbb{R}^{N \times q}. \quad (7)$$

Remark 1. In Eq. (7), N denotes the graph node count—the row dimension of the eigenvector matrix $\mathbf{U}_{1:q} \in \mathbb{R}^{N \times q}$ —and is distinct from the input feature dimension D used in Section 2.2. For the PPCA loading matrix $\mathbf{W}_{PPCA} \in \mathbb{R}^{D \times q}$ the row index ranges over feature coordinates; here it ranges over graph nodes. In CORA, for example, $N = 2708$ (nodes) while $D = 1433$ (bag-of-words feature dimension). The spectral loading matrix therefore has shape $(2708 \times q)$, projecting from the q -dimensional eigenbasis onto all N graph nodes—not onto feature coordinates.

Equation (7) is the central structural commitment of the VDT: the loading matrix lives in the Laplacian eigenbasis. High-frequency directions are suppressed both by the mode weight prior (Section 4.4) and by the eigenvalue-weighted prior on $\mathbf{S}$ (Section 4.3). The independence of $\log \sigma_{\mathbf{S}}^2$ from $\mathbf{S}$ is a deliberate design decision (fix #52): the old proxy $\log(\mathbf{S}^2 + \epsilon)$ conflated posterior mean and variance, producing biased KL gradients.

3.4 DifferentiableLaplacian

The loading matrix $\mathbf{W} \in \mathbb{R}^{N \times q}$ is converted to a graph Laplacian by:

$$w_{ij}(\mathbf{W}) = w_{ij}^{\text{base}} \cdot \sigma(-\|\mathbf{W}_i - \mathbf{W}_j\|^2), \quad \mathbf{L}(\mathbf{z}) = \mathbf{D}(\mathbf{W}) - \mathbf{A}(\mathbf{W}), \quad (8)$$

where σ is the logistic sigmoid, w_{ij}^{base} is a frozen base-graph weight, and $\mathbf{W}_i \in \mathbb{R}^q$ denotes the i -th row of $\mathbf{W}$ (the spectral embedding of node i). Since $\|\mathbf{W}_i - \mathbf{W}_j\|^2 \geq 0$ by definition, $\sigma(-\|\mathbf{W}_i - \mathbf{W}_j\|^2) \in (0, 0.5]$, so all edge weights are *strictly positive* and lie in $(0, w_{ij}^{\text{base}}/2]$. The construction remains fully differentiable through $\mathbf{W}$.

Quantum-like interference in the heat kernel does not arise here from negative edge weights. Instead, it arises from the *signed structure* of $\mathbf{W}$: because the entries of $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$ can be positive or negative, rows $\mathbf{W}_i$ and $\mathbf{W}_j$ may be positively or negatively correlated in the eigenvector subspace. These signed correlations reshape the eigenvector decomposition of $\mathbf{L}(\mathbf{z})$: modes whose eigenvectors align with anti-correlated row pairs carry larger Laplacian eigenvalues and are exponentially suppressed in the heat kernel $\mathbf{K}_\tau = \mathbf{U} \exp(-t\mathbf{\Lambda}) \mathbf{U}^\top$, producing constructive or destructive-like interference in the diffusion pathways between nodes—the real-amplitude counterpart of complex phase cancellation [2].

3.5 DiffusionDecoder (TauModeDiffusion)

The decoder applies a truncated heat kernel to the embedding table $\mathbf{E}$ and returns the heat-diffused row for each queried node:

$$\mathbf{K}_\tau = \mathbf{U}_\tau \exp(-\tau \mathbf{\Lambda}_\tau) \mathbf{U}_\tau^\top, \quad \hat{\mathbf{x}} = (\mathbf{K}_\tau \mathbf{E})_{[\text{node_idx}]}, \quad (9)$$

where the summation is truncated to the leading τ_{modes} eigenpairs of $\mathbf{L}(\mathbf{z})$, and $\mathbf{E} \in \mathbb{R}^{N \times D}$ is the frozen node embedding table. The reconstruction is purely spectral: no MLP residual correction is applied inside `TauModeDiffusion` or in `WiringAutoencoder.forward()` after the diffusion step. The heat kernel $\mathbf{K}_\tau$ is a real symmetric positive-semi-definite operator—the real-valued analogue of a quantum channel [2, 10, 17].

4 The Four-Term ELBO

The VDT objective is the negative ELBO:

$$\mathcal{L}_{\text{VDT}} = \underbrace{-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}, \mathbf{W})]}_{\text{NLL recon}} + \underbrace{D_{\text{KL}}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\text{latent KL}} + \underbrace{D_{\text{KL}}(q(\mathbf{S}) \parallel p(\mathbf{S}|\mathcal{I}))}_{\text{spectral basis KL}} + \underbrace{D_{\text{KL}}(q(\boldsymbol{\omega}) \parallel p(\boldsymbol{\omega}|\tau, \boldsymbol{\Lambda}))}_{\text{taumode KL}} \quad (10)$$

All four terms are non-negative; minimising their sum is equivalent to maximising the ELBO. We derive each term below.

4.1 Term 1: Reconstruction

Under the Gaussian output model the reconstruction term is:

$$-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}, \mathbf{W})] = \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|^2}{2\sigma^2} + \frac{D}{2} \log(2\pi\sigma^2). \quad (11)$$

In the implementation the constant $\frac{D}{2} \log(2\pi)$ is dropped; σ is a learnable scalar log-variance.

4.2 Term 2: Isotropic Latent KL

The standard VAE isotropic KL [8]:

$$D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) = -\frac{1}{2} \sum_{j=1}^{d_z} \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right). \quad (12)$$

4.3 Term 3: Spectral-Basis KL

The coefficient matrix $\mathbf{S} \in \mathbb{R}^{q \times q}$ receives a *row-independent* Gaussian prior in which each entry S_{kj} is penalised by the k -th eigenvalue of the frozen Laplacian index:

$$p(S_{kj} \mid \mathcal{I}) = \mathcal{N}\left(0, \frac{1}{\lambda_S \lambda_k}\right) \quad \forall k \in [q], j \in [q]. \quad (13)$$

Because the prior factorises over both indices, the joint log-density can be written as the *trace form*:

$$\log p(\mathbf{S} \mid \mathcal{I}) \propto -\frac{\lambda_S}{2} \text{tr}(\mathbf{S}^\top \boldsymbol{\Lambda}_{1:q} \mathbf{S}) = -\frac{\lambda_S}{2} \sum_{k=1}^q \lambda_k \|\mathbf{S}_k\|^2, \quad (14)$$

where $\boldsymbol{\Lambda}_{1:q} = \text{diag}(\lambda_1, \dots, \lambda_q)$ and $\mathbf{S}_k \in \mathbb{R}^q$ is the k -th row of $\mathbf{S}$. Large eigenvalues penalise the corresponding rows of $\mathbf{S}$ more strongly—a *spectral Occam’s razor*: high-frequency components of the loading matrix are shrunk towards zero.

The variational posterior is $q(\mathbf{S}) = \mathcal{N}(\mathbf{S}_0, \text{diag}(\exp(\log \boldsymbol{\sigma}_{\mathbf{S}}^2)))$, where $\mathbf{S}_0$ and $\log \boldsymbol{\sigma}_{\mathbf{S}}^2$ are *independent* outputs of `SpectralLoadingDecoder` (fix #52). Under Gaussian–Gaussian conjugacy the KL is closed-form.

Remark 2. The prior in Eq. (13) is row-independent with identity column precision ($\boldsymbol{\Lambda}_{\text{col}} = \mathbf{I}$). A full matrix-normal prior would have the form $\log p(\mathbf{S}) \propto -\frac{1}{2} \text{tr}(\mathbf{S}^\top \boldsymbol{\Lambda}_{\text{row}} \mathbf{S} \boldsymbol{\Lambda}_{\text{col}})$ with two separate precision matrices. When $\boldsymbol{\Lambda}_{\text{col}} = \mathbf{I}$ this reduces exactly to Eq. (14). The implementation in `vdtspectral.py::spectral_basis_kl` broadcasts the precision vector `prec = lambda_S * eigvals_q` over the column axis of $\mathbf{S}$, which is the direct computational realisation of Eq. (13): one scalar precision per row k , shared across all q columns j .

4.4 Term 4: taumode Frequency KL

Mode weights $\omega_k > 0$ receive an exponential prior indexed by frequency:

$$p(\omega_k | \tau, \lambda_k) = \text{Exponential}(\tau \lambda_k), \quad (15)$$

giving heavy support on low-frequency modes (small λ_k). The variational posterior is a Gamma: $q(\omega_k) = \text{Gamma}(a_k, b_k)$. The KL between a Gamma posterior and an Exponential prior is closed-form. Setting $r = \tau \lambda_k$ (prior rate), the derivation proceeds as:

$$\begin{aligned} \text{KL}(\text{Gamma}(a, b) \parallel \text{Exp}(r)) &= \mathbb{E}_q[\log q(\omega) - \log p(\omega)] \\ &= -H[\text{Gamma}(a, b)] - \log r + r \mathbb{E}_q[\omega] \end{aligned}$$

where the Gamma entropy is $H[\text{Gamma}(a, b)] = a - \log b + \ln \Gamma(a) + (1 - a)\psi(a)$ and $\mathbb{E}_q[\omega] = a/b$. Substituting and simplifying gives:

$$D_{\text{KL}}(\text{Gamma}(a, b) \parallel \text{Exp}(r)) = \log b - \log r - \ln \Gamma(a) - (1 - a)\psi(a) - a + \frac{ar}{b}, \quad (16)$$

where ψ is the digamma function and $r = \tau \lambda_k$. No Monte Carlo sampling is required. The implementation in `vdt/spectral.py::tau_mode_kl` follows Eq. (16) exactly; note the *negative* sign on $\ln \Gamma(a)$ and the additional $-a$ term, both of which arise from expanding the Gamma entropy.

Remark 3. A common sign error writes $+\ln \Gamma(a)$ instead of $-\ln \Gamma(a)$. The correct sign follows immediately from the entropy expansion: $H = a - \log b + \ln \Gamma(a) + (1 - a)\psi(a)$, which enters the KL with a negative prefactor ($\text{KL} = -H + \dots$), flipping all signs, including the sign on $\ln \Gamma(a)$.

4.5 Comparison with the Base Wiring AE

Table 1: ELBO structure: Base Wiring AE vs. VDT.

Term	Wiring AE (base)	VDT
Reconstruction	$-\log p(\mathbf{x} \mathbf{z})$	Same
Latent KL	$D_{\text{KL}}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))$ isotropic	Same
Spectral basis	None	$D_{\text{KL}}(q(\mathbf{S}) \parallel p(\mathbf{S} \mathcal{I}))$ eigenvalue-weighted
Frequency	αJ_{freq} hard penalty	$D_{\text{KL}}(q(\boldsymbol{\omega}) \parallel \text{Exp}(\tau \lambda_k))$ variational

5 Spectral Artefact and Associative Memory

After training, the model operates in a two-phase regime.

5.1 Phase 1: Spectral Artefact Extraction

At the prior mean ($\mathbf{z} = \mathbf{0}$) the decoder emits a *spectral artefact*:

$$\mathcal{A}(\mathcal{I}) = \{\hat{\mathbf{W}}, \hat{\boldsymbol{\omega}}, \mathbf{S}_{\text{mem}}\}, \quad \mathbf{S}_{\text{mem}} = \sum_{k=1}^q \hat{\omega}_k d_{\theta}(\hat{\mathbf{w}}_k) \hat{\mathbf{w}}_k^{\top}, \quad (17)$$

where $\hat{\mathbf{w}}_k \in \mathbb{R}^N$ is the k -th column of $\hat{\mathbf{W}} \in \mathbb{R}^{N \times q}$ and $d_{\theta}(\hat{\mathbf{w}}_k)$ is the decoder response at spectral direction k . $\mathbf{S}_{\text{mem}}$ is a Hopfield outer-product memory matrix [16].

5.2 Phase 2: Spectral Memory Transformer

The transformer feed-forward and cross-attention value matrices are initialised from $\mathbf{S}_{\text{mem}}$, providing *long-term spectral prior memory*. Online delta-rule updates $\mathbf{S}_{\text{mem}} \leftarrow \mathbf{S}_{\text{mem}} + \eta(\mathbf{v} - \mathbf{S}_{\text{mem}}\mathbf{k})\mathbf{k}^\top$ write new associations without corrupting spectral key orthogonality. Key orthogonality (inherited from Laplacian eigenvectors) maximises retrieval SNR by construction.

5.3 Entropy-Controlled Generation

The spectral entropy of the generated Laplacian [18]:

$$H(\mathbf{L}(\mathbf{z})) = - \sum_{k=1}^N \bar{\lambda}_k \log \bar{\lambda}_k, \quad \bar{\lambda}_k = \frac{\lambda_k}{\sum_j \lambda_j}, \quad (18)$$

can be steered by conditioning $\mathbf{z}$ on a target entropy level, enabling *entropy-controlled spectral graph generation*—a capability absent in flat-prior VAEs.

6 Stability Analysis

6.1 Six-Level Hierarchy

Training stability is analysed through a six-level hierarchy [4]; failures at lower levels must be resolved before diagnosing higher ones:

1. **Graph geometry.** $\lambda_{\max}(\mathbf{L}_f)$ finite; CFL condition satisfied; graph connectivity.
2. **Wave update per mode.** Damping coefficients $\gamma_k > 0$; underdamped/overdamped classification.
3. **Recurrent VDT dynamics.** Energy $\mathcal{E}_t$ monotone; spectral entropy H_{spectral} stable.
4. **Density matrix.** $\boldsymbol{\rho}_t^+$ positive semi-definite; $\|\boldsymbol{\rho}_t\|_F$ bounded.
5. **Optimiser.** Preconditioned gradient descent convergence: $\kappa(\mathbf{P}_{\sigma,M}) \ll \kappa(\mathbf{L}_f)$; step size $\eta < 2/L_{\sigma,M}$.
6. **Mode weights.** Active mode count $N_{\text{active}} = |\{k : \mathbb{E}[\omega_k] > \delta\}| \geq q_{\min}$; $\|\hat{\mathbf{W}}\|_F$ non-degenerate.

6.2 Mode Weight Collapse

For high-frequency modes with large $\tau\lambda_k$, the Exponential prior pushes $q(\omega_k)$ towards a point mass near zero. This is *correct behaviour* (mode selection), but if it extends to low-frequency modes the loading matrix degenerates. Two mitigations are applied:

- **Shape-parameter floor:** $a_k \geq a_{\min} = 0.1$ prevents full collapse.
- **Active-mode penalty:** a soft Lagrange term $\nu \cdot \text{relu}(q_{\min} - N_{\text{active}})$ maintains a minimum number of active modes.

6.3 Spectral Eigenvalue Floor

When the Fiedler eigenvalue $\lambda_2 \approx 0$ (nearly disconnected graph) the spectral-basis KL prior becomes nearly flat and produces high gradient variance. A floor applied in the KL computation only ($\lambda_k \leftarrow \max(\lambda_k, 10^{-3})$) prevents numerical instability without affecting the spectral geometry.

7 Experiments

7.1 Datasets

We evaluate on three benchmarks:

- **CORA** [12]: 2708 nodes, 5429 edges, 7 classes, 1433-dim bag-of-words features.
- **PubMed** [12]: 19717 nodes, 44338 edges, 3 classes.
- **Synthetic spring networks**: 400 randomly generated spring graphs with controllable spectral entropy.
- **Open Graph Benchmark (OGB)** [19]: standardised large-scale graph datasets used for held-out evaluation.

7.2 Evaluation Metrics

Table 2: VDT benchmark metrics.

Metric	Symbol	Measures
Reconstruction MSE	$\ \mathbf{x} - \hat{\mathbf{x}}\ ^2$	Wiring path fidelity
Latent KL	KL_z	Isotropic regularisation
Spectral basis KL	KL_S	Alignment with index $\mathcal{I}$
taumode KL	KL_τ	Frequency band selection
Active modes	N_{active}	Modes with $\mathbb{E}[\omega_k] > 0.01$
Memory SNR	d_k/N_{stored}	Hopfield retrieval quality
ELBO Bayes factor	$\exp(\mathcal{L}(\mathcal{I}_1) - \mathcal{L}(\mathcal{I}_2))$	Index comparison
Linear probe acc.	$\text{Acc}_{\text{probe}}$	Latent discriminability

7.3 Flagship Demo: Entropy-Controlled Spectral Generation

Standard VAEs decode $\mathbf{z}$ into flat feature vectors; the VDT decodes $\mathbf{z}$ into a *graph wiring*—a Laplacian whose eigenvalues are vibrational modes of the spring network (cf. Rayleigh [1]). The latent space directly encodes spectral geometry, enabling samples whose spectral entropy matches a target level. This capability is unique to the VDT’s vibrational latent representation and is unavailable in flat-prior models.

8 Related Work

Spectral graph autoencoders. Graph variational autoencoders [12] encode topology in the latent space but do not use spectral priors. VGAE and its successors decode adjacency matrices directly, without a vibrational interpretation.

Probabilistic PCA on graphs. Laplacian Eigenmaps [10] and Diffusion Maps [11] use the heat kernel for dimensionality reduction but are deterministic and lack variational objectives.

NVIB. The Nonparametric Variational Information Bottleneck [15] applies variational compression to sequence models. The VDT shares the ELBO philosophy but applies spectral-structural priors rather than capacity constraints on a fixed-dimension latent.

Beta-VAE and disentanglement. Higgins et al. [9] introduce β -VAE, showing that stronger KL regularisation encourages disentangled latent representations. The VDT’s eigenvalue-weighted spectral-basis KL can be seen as an anisotropic generalisation of this pressure, tuning the regularisation strength independently per spectral frequency band.

Real quantum mechanics. Renou et al. [13] and McKague et al. [14] provide experimental and theoretical grounding for real-valued quantum theories. The VDT does not claim to *be* a quantum system, but borrows the structure of signed-amplitude interference as a computational primitive.

9 Discussion and Future Work

The VDT establishes a formal connection between three bodies of theory: Rayleigh’s vibrational mechanics, Aaronson’s real-amplitude quantum probabilities, and Bayesian graph-spectral learning. The central contribution is the four-term ELBO (10), which replaces hard spectral penalties with proper variational counterparts while preserving the ArrowSpace wiring geometry.

A Spectral Diffusion improvement on this architecture is work in progress.

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```

Input x (B, D)          Frozen: U_q (N, q),  eigvals_q (q,)
|                        |
+----[fingerprint from eigvals_q, derived internally]----+
|
v
+-----+
| WiringEncoder          |
| MLP + eigenvalue fingerprint |
| --> mu, log_var        |
|   [reparameterise --> z] |
| --> log_a, log_b        |
|   [ModeWeightHead --> q(w)] |
+-----+
|
z_to_q: z (B, latent_dim) --> z_q (B, q)
|
v
+-----+
| SpectralLoadingDecoder |
| inputs: z_q, U_q, L_base |
| outputs: W, omega, S, log_var_S, L_z |
| W = U_q @ diag(omega) @ S (B, N, q) |
+-----+
|           |           |
|   W       | (S, log_var_S) | (log_a, log_b)
|           |           |
| L_z (B,N,N) | kl_S         | kl_tau
|           |           |
| TauModeDiffusion(L_z, E, node_idx)
|
x_hat --> recon_loss

loss = recon + kl_z + kl_S + kl_tau

```

Figure 1: VDT v2 data flow (forward signature: $(x, U_q, \text{eigvals}_q, \text{node_idx}, \text{spectral_cache})$). The index $\mathcal{I}$ enters only through the frozen eigenpair (U_q, λ_q) precomputed once before training; no Laplacian eigendecomposition is performed at training time. The eigenvalue fingerprint is derived *internally* by `WiringEncoder` from `eigvals_q`. `log_var_S` is a separate output of `SpectralLoadingDecoder` required for the KL_S computation (fix #52). Note: N is the graph node count; D is the input feature dimension. These differ in practice (e.g. CORA: $N=2708$, $D=1433$).